

Commissioning of the Vera C. Rubin Observatory and Weak Gravitational Lensing

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The Vera C. Rubin Observatory began commissioning its camera, LSSTCam, in April 2025, with the Legacy Survey of Space and Time (LSST) scheduled to start in 2026. A primary science goal is constraining Dark Energy through weak gravitational lensing of the large-scale structure (cosmic shear). After a full year of data from LSST, these measurements are expected to reach precision comparable to recent Dark Energy Spectroscopic Instrument (DESI), providing an independent test of hints that Dark Energy may evolve over time. However, cosmic shear requires exquisite control of instrumental systematics. This proceeding presents an overview of the Rubin Observatory commissioning — the successes achieved and the systematic issues we are working to resolve.

1 The Vera C. Rubin Observatory

The Vera C. Rubin Observatory is an 8.4-meter aperture optical wide-field survey telescope located at Cerro Pachón (2,682 m) in Chile (Ivezić et al. 2019¹). Commissioning began in late 2024 with a commissioning camera, LSSTComCam (0.5 deg²), whose data were released as Data Preview 1 (DP1) in mid-2025 (the Vera C. Rubin Observatory Team 2026²). The main camera, LSSTCam (9.6 deg²), started commissioning in April 2025 and has been operating ever since. The first batch of LSSTCam data is expected to be published as Data Preview 2 (DP2) (Guy et al. 2025³) in mid-2026, with real-time alerts having begun in February 2026. The Legacy Survey of Space and Time (LSST) is scheduled to start in 2026.

The Rubin Observatory has broad scientific objectives spanning from solar system physics to constraining the properties of Dark Energy. On this last topic, the Rubin Observatory is expected to constrain dark energy properties through multiple cosmological probes. Among them, the 3×2 pt analysis, combining cosmic shear, galaxy-galaxy lensing, and galaxy clustering, is expected to reach the best precision alone. It should reach a precision comparable to the current combination of DESI and CMB results after just one year of the LSST survey — provided systematic effects are under control. However, early LSSTCam data show this is more challenging than anticipated.

In this proceeding, we discuss how image quality and Point Spread Function (PSF) modeling challenge cosmic shear measurements, what we discovered once on sky, how we have improved,

and what remains to be done. Despite these challenges, we show that we can already detect a weak lensing signal with no obvious PSF contamination in the case of galaxy cluster lensing, which has less strict requirements than cosmic shear.

2 On PSF modeling and image quality

The modeling of the PSF is one of many systematic effects that can affect cosmic shear measurements (see, e.g., Mandelbaum 2018⁴). Recent Dark Energy constraints have included dedicated analyses to study the impact of PSF systematics on 3×2 pt measurements (Schutt et al. 2025⁵, Yamamoto et al. 2025⁶). The general principle is to achieve an accurate description of second and higher order moments (Zhang et al. 2023⁷, Schutt et al. 2025⁵) and to minimize their correlations in sky coordinates (Schutt et al. 2025⁵). Even when the PSF model is well described, image quality remains an important metric. Visits with poor image quality decrease survey depth, thereby raising the detection threshold for faint sources such as high-redshift galaxies, decrease the number of resolved galaxies, and overall reduce the statistical power of the survey (Dark Energy Science Collaboration (DESC) SRD 2018⁸).

The Rubin Observatory PSF model is built within the LSST Science Pipelines (The Rubin Observatory Science Pipelines Team 2025⁹) using PSFs In the Full Field of view (PIFF; Jarvis et al. 2021¹⁰) and is based on the same framework used for the Dark Energy Survey (DES) Y6 PSF model. During commissioning, the main limitation to image quality was optical aberrations, which are controlled by the Active Optics System (AOS) described in Megias Homar et al. 2024¹¹. Both PSF modeling and image quality during the commissioning phase did not perform as expected for the LSST survey. The following sections present a walkthrough of the problems encountered, how we addressed them, what remains challenging, and upcoming improvements to PSF modeling and image quality.

3 On improving PSF and image quality

The data presented here are LSSTCam observations from the period that will form the basis of Data Preview 2, which was not yet public at the time of writing but will be released in mid-2026. It corresponds to images acquired between April 2025 and January 2026, totaling approximately 30,000 single-visit exposures across the six filters of the Rubin Observatory. These data are used throughout this proceeding to evaluate both PSF modeling and, indirectly, image quality through PSF residuals.

3.1 PSF residuals in focal plane coordinates

A standard way to quantify biases in PSF modeling is by examining averages of PSF second moment residuals in focal plane coordinates, as done for example in DES Y6 (Schutt et al. 2025⁵). Here we focus on residuals of the trace of the second moment matrix T , defined as:

$$\frac{\delta T}{T} = \frac{T_{\text{star}} - T_{\text{PSF}}}{T_{\text{star}}} \quad (1)$$

where T_{star} is measured on stars used for training or validating the PSF model, and T_{PSF} is estimated directly from the PSF model. DESC has strict requirements on $\delta T/T$: below 0.4% for LSST Year 1 and below 0.1% for LSST Year 10 (DESC SRD 2018⁸). In the following, we focus mostly on $\delta T/T$, but ellipticity residuals exhibit similar patterns. The analysis presented in this section and the next is based on $\sim 3,000$ visits, which was approximately what was available after full pipeline processing in the pre-DP2 era.

Figure 1 shows $\delta T/T$ averaged over these visits and projected into focal plane coordinates. The focal plane is paved with two distinct types of Charge-Coupled Devices (CCDs) from different manufacturers: E2V sensors in the center and ITL sensors on the edges. Their locations

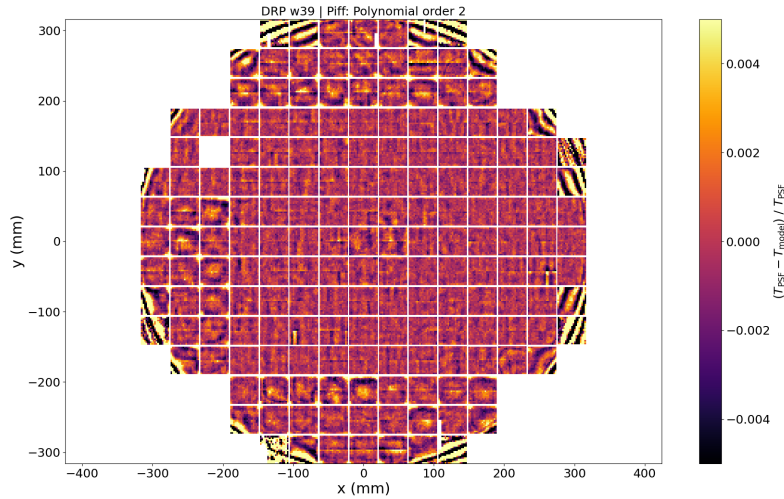


Figure 1 – $\delta T/T$ projected in focal plane coordinates for a subset of Data Preview 2. No image quality cut is applied. The PSF model uses a 2nd order polynomial interpolation.

are easily distinguishable as they exhibit different features. Three main patterns stand out: an ITL “blob”, amplifier offsets on the E2V sensors, and a ring pattern at the edge of the focal plane. While a detailed investigation of all three features is possible, this proceeding focuses primarily on the ITL blob, which reveals information about both PSF model performance and, unexpectedly, the contribution of the optics to our image quality budget. See the next section for details.

3.2 On the ITL “blob” or is my telescope in focus?

This ITL blob in the $\delta T/T$ map was already observed with the commissioning camera and described in the DP1 paper (Rubin Observatory Team 2026²). The solution adopted then, shown in Figure 16 of the DP1 paper, was to allow more flexibility in the spatial variation of the PSF model across a CCD by using a 4th order polynomial interpolation instead of the standard 2nd order. As seen in that figure, this removes the ITL blob, but it does not explain the physics behind this pattern that appears only in ITL sensors.

To better understand the origin of this feature, instead of increasing to a 4th order polynomial, we switch to a 0th order polynomial, as shown in Figure 2. A clear difference remains between E2V and ITL sensors: the ITL sensors exhibit a curved circular pattern, while the E2V sensors show a tilt-like shape. If we interpret this as local variations in focus across the camera due to height variations or non-flatness of the focal plane, it should simply reflect the physical shape of the sensor array — and indeed, this is what we observe. Figure 2 shows the height map of the camera as measured in the lab at SLAC National Accelerator Laboratory (SLAC) (Roodman et al. 2024¹²). It is straightforward to see that $\delta T/T$ is highly correlated with the height map; therefore, the ITL blob is due to the complex surface shape of the sensors themselves.

However, why is this happening at all? When the height measurements were made at SLAC, this was not a concern, so why is the height map now imprinted on our PSF residuals? The answer is that the telescope was mostly out of focus until mid-December 2025. The Active Optics System (AOS) was slightly off-target: it was reaching the focus value requested, but that target was not the true optimal focus. This was corrected in December 2025, but by then most of the data destined for DP2 had already been acquired slightly out of focus. This explains why the 4th order polynomial interpolation was needed to remove the blob. Further analysis is required to determine whether this higher-order interpolation is still necessary for the on-sky

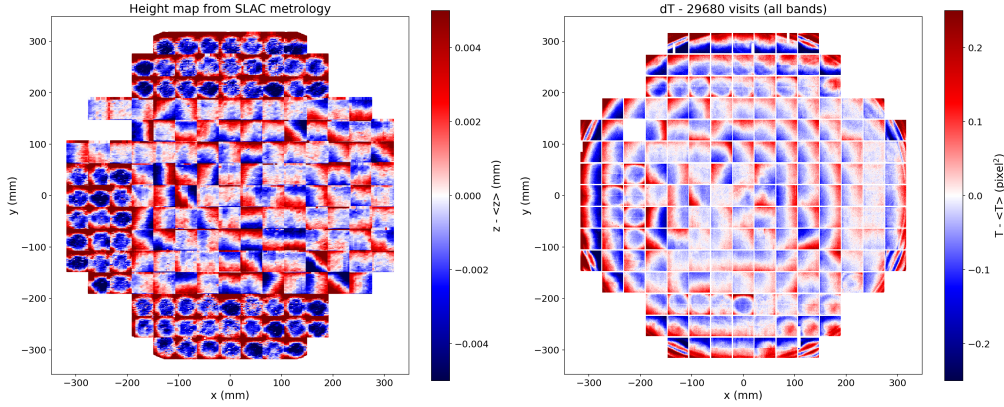


Figure 2 – Left: Height map of LSSTCam measured at SLAC. Right: PSF size residuals measured on thousands of science images before any interpolation is done (mean interpolation). Both Left and right match.

data currently being gathered with the corrected focus settings.

3.3 PSF residuals in sky coordinates

Switching to a 4th order polynomial interpolation helps remove most of the features in the $\delta T/T$ maps, as demonstrated in the DP1 paper (Rubin Observatory Team 2026²). However, most analyses during the first year of the Rubin Observatory commissioning were performed on a limited subsample of visits. The Rubin Observatory Data Management began processing DP2 data in January 2026, with PSF fitting completed during the same period. This represented an order of magnitude increase in data volume, totaling approximately 30,000 visits. With this larger dataset, we were able to perform more detailed band-specific analyses. One notable finding was a global offset in $\delta T/T$ that depended on the observing band.

To understand the origin of this offset, we examine the spatial distribution of $\delta T/T$ across the sky. Figure 3 shows a sky map of $\delta T/T$ along with the positions of the Milky Way, the Large Magellanic Cloud (LMC), and the Small Magellanic Cloud (SMC). A clear correlation with stellar density emerges: there is a larger bias in these high-density regions. The offset therefore depends on how much time the survey spent observing high stellar density fields in each band. We attribute this offset to problems in sky subtraction. Areas with more stars are more challenging for sky estimation, and an incorrect background estimate biases second moment measurements at low signal-to-noise and the PSF model at high signal-to-noise. We have developed a proof of concept to address this issue in PSF modeling, but it will not be available for Data Preview 2. It should, however, be implemented for the first data release of LSST.

3.4 Modeling the chromaticity of the PSF

Differential chromatic refraction (DCR) and atmospheric turbulence introduce a chromatic dependence of the PSF that manifests as a “fatter-bluer” effect. This can be seen in Figure 4 when not accounted for. Unlike the Hyper Suprime-Cam (HSC), LSSTCam does not have an Atmospheric Dispersion Corrector, so this effect must be modeled explicitly. In DES Y6 (Schutt et al. 2025⁵), the approach was to add a third axis of interpolation for the PSF model coefficients, where this third axis is stellar color. We re-implemented this DES Y6 correction in the Rubin Observatory Data Management pipeline. Figure 4 shows the performance of this correction on Data Preview 1 data, which appears comparable to DES Y6. This may be sufficient for LSST Year 1, but as suggested in Plazas et al. 2012¹³ and Meyers et al. 2015¹⁴, it is likely insufficient for LSST Year 10 and may require more sophisticated modeling.

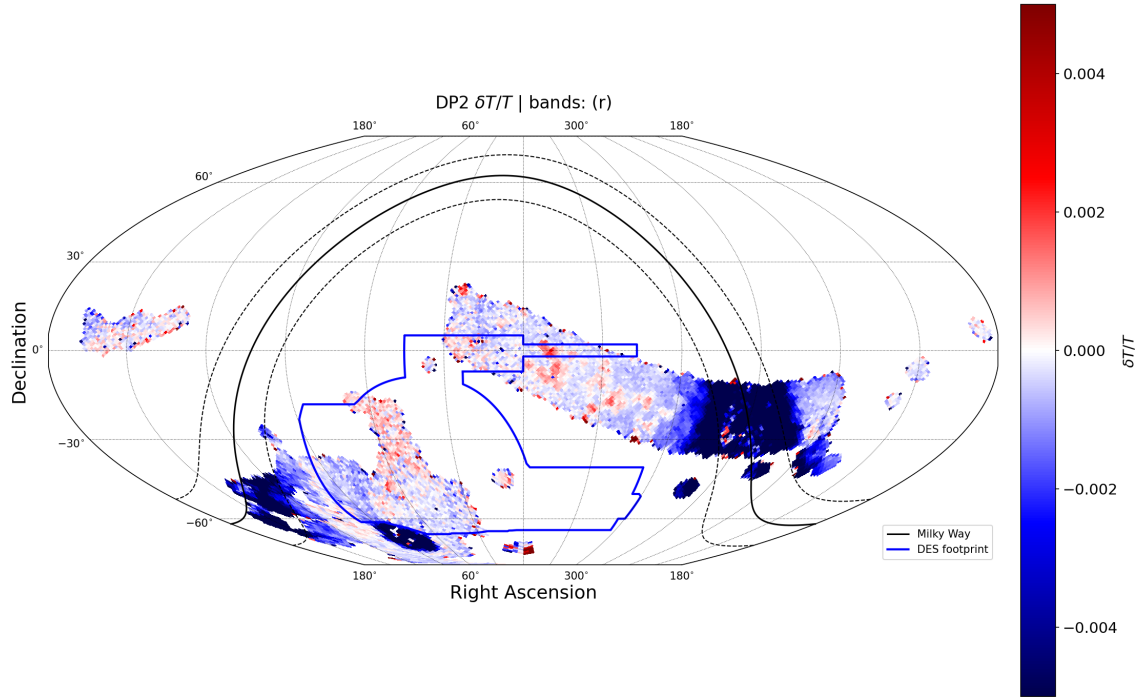


Figure 3 – $\delta T/T$ projected in sky coordinates for single visits in Data Preview 2. No image quality cut is applied.

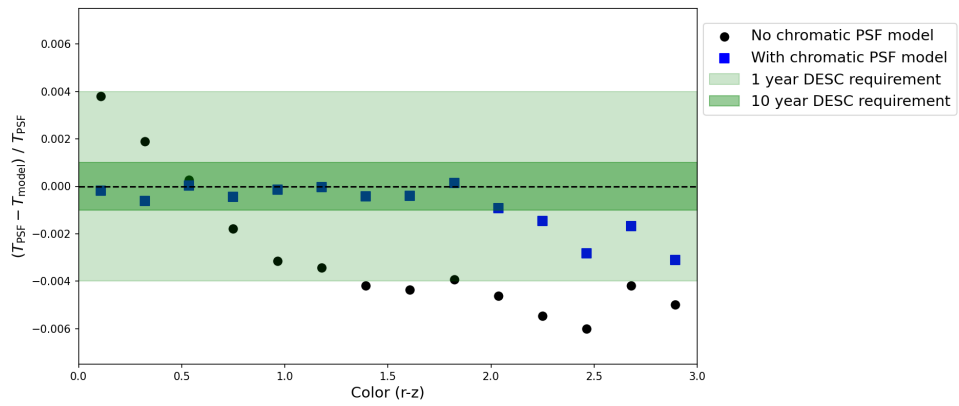


Figure 4 – $\delta T/T$ as a function of $r - z$ in Data Preview 1, with and without chromatic correction.

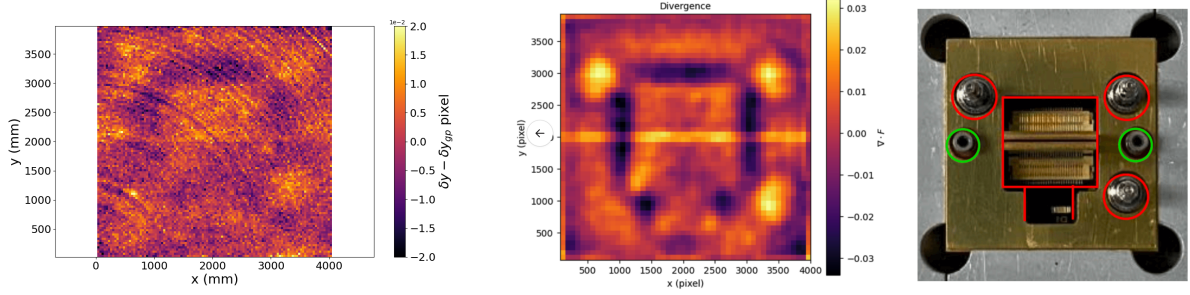


Figure 5 – Left: Stack of astrometric residuals in a single direction (y direction of the CCD). Two patterns are observed: the known and expected tree rings, but also a smoother pattern with slower variation. Center: By averaging all the sensors together and by taking the divergence of the field, a clear structure appears that looks to be exactly the shape of the support of the detector (Right). This is for one of the 2 types of sensors that are on the focal plane (ITL), and the other type (E2V) exhibits similar structures.

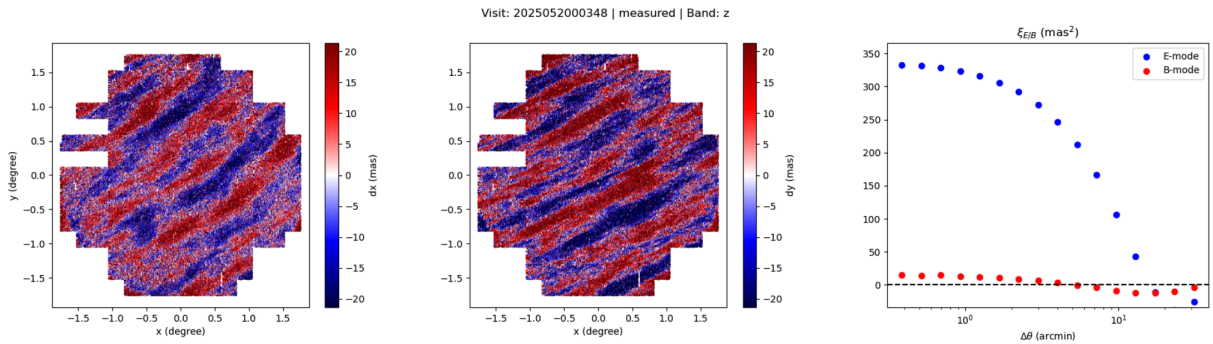


Figure 6 – Astrometric residuals from a single visit of LSSTCam. The two first panels on the left show each component of the astrometric residuals field and the last panel shows the E and B mode for this visit. The stripe patterns, which are pure E-mode, are characteristic of atmospheric turbulence and significantly higher than what was observed in previous surveys.

3.5 Improving astrometry for PSF

Another component of PSF modeling inspired by the Dark Energy Survey is working in sky coordinates. One advantage of PIFF is that it can operate in sky coordinates, thereby folding spatial variations due to CCD defects such as tree rings (Plazas et al. 2014¹⁵) into the World Coordinate System (WCS) transformation. This was one of the main reasons for PIFF’s success in the DES cosmic shear analysis. The astrometry in the Rubin Observatory is modeled using the same approach as DES (Bernstein et al. 2017¹⁶). In addition to characterizing CCD defects, as shown in Figure 5, we implemented a correction for atmospheric turbulence.

In DES and HSC, it was shown that astrometric residuals are dominated by atmospheric turbulence (Fortino et al. 2021¹⁷, Gomes et al. 2025¹⁸, Léget et al. 2021¹⁹). A key difference with the Rubin Observatory compared to DES and HSC is the 30-second exposure time, which is significantly shorter than both surveys. The direct implication is that the variance is at least twice as large in the Rubin Observatory as in DES due to the shorter integration time. However, as demonstrated in DES and HSC, an anisotropic Gaussian Process (GP) can be used to correct for this effect. We applied the same method as HSC (Léget et al. 2021¹⁹), and as in HSC and DES, we are able to remove more than 90% of the E-mode power (see Figures 6 and 7). This correction is part of our full WCS solution but is not yet used in PSF modeling for Data Preview 2; it will be incorporated in the first year data release.

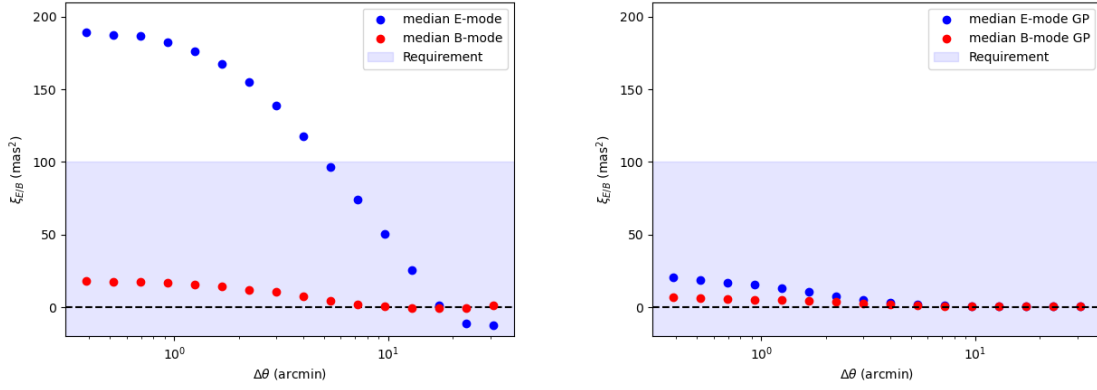


Figure 7 – Left: average across visits of E- and B- mode of the astrometric residuals field before GP modeling and on the right after GP modeling. GP removes more than 90% of the observed variance due to the atmospheric turbulence.

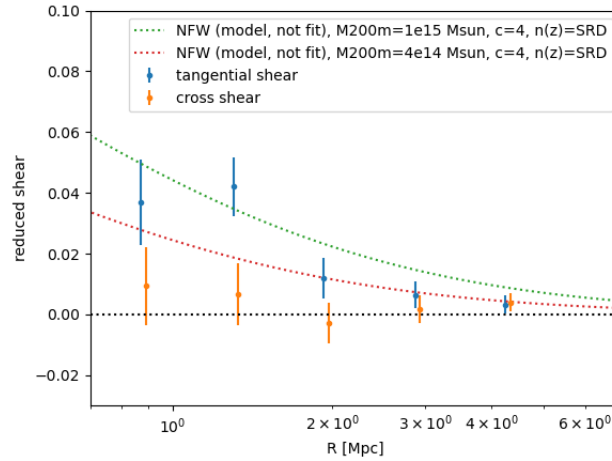


Figure 8 – Tangential and cross-shear profiles of PSZ2 G309.43-72.86 measured on LSSTCam data from Data Preview 2.

4 First detection of weak lensing signal with LSSTCam

Commissioning has revealed both expected and unexpected features. As this proceeding demonstrates, significant progress has been made, and improvements continue as we refine our understanding of the instrument. A natural question arises: will it still be possible to extract a weak lensing signal from Data Preview 2, scheduled for mid-2026? In the spirit of the Data Preview 1 analysis of the galaxy cluster Abell 360 (Combey et al. 2025²⁰), based on the Rubin tutorials (Rubin Observatory 2025²¹), we perform a weak lensing detection on a known cluster to check for potential contamination from PSF residuals. By cross-correlating the Data Preview 2 footprint with the Sunyaev-Zel’dovich (SZ) catalogue from Planck, we identified one cluster in a deep drilling field (ELAIS-S1) that serves as a good candidate: PSZ2 G309.43-72.86, at a redshift of $z = 0.35$ with an SZ mass of $5.82 \times 10^{14} M_{\odot}$. We applied the same selection criteria as described in Combey et al. 2025²⁰ and measured the tangential shear and cross-shear signals for PSZ2 G309.43-72.86, shown in Figure 8. As expected for the weak lensing signal of a galaxy cluster, the tangential shear signal is non-zero while the cross-shear signal is consistent with zero. This provides evidence that the Rubin Observatory pipeline can deliver data products suitable for cosmological science, even at the Data Preview 2 stage. With the public release of DP2 in mid-2026, further validation and science results will follow.

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